

Sales Performance and Product Demand Analysis for Improved Inventory Planning in a Retail Garments Store

A Proposal report for theBDM capstone Project

Submitted by :

Name: Sumaiya Afroz

Roll number: 24f1000754@ds.study.iitm.ac.in



**IITM Online BS Degree Program,
Indian Institute of Technology,
Madras, Chennai Tamil Nadu, India,**

600036

Contents

Declaration Statement	2
1 Executive Summary	3
2 OrganizationBackground	3
3 Problem Statement	4
3.1 Problem 1: Optimization of Locked Working Capital	
3.2 Problem 2 : Minimization of Total Inventory Cost	
4 Background of the Problem	4
5 Problem Solving Approach	5-6
6 Expected Timeline	6
6.1 Work Breakdown Structure	6
6.2 Gantt Chart	7
7 Expected Outcome	7

Declaration Statement

I am working on a Project Title “Sales Performance and Product Demand Analysis for Improved Inventory Planning in a Retail Garments Store”. I extend my appreciation to Matching Point, for providing the necessary resources that enabled me to conduct my project. I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability. Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures. I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report. I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority. I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.



Signature of Candidate

Name: Sumaiya Afroz

Date: 12-07-26

1 Executive Summary

Matching Point is a retail clothing and household utility store located on Powerhouse Road, Maripur, Muzaffarpur, Bihar, established on 19 May 2014. The store is a B2C business that has been serving local customers for over ten years. It offers a wide range of products including lungi, undergarments, sarees, blouses, suit pieces, matching dupattas, saaya (petticoat), leggings, plazo, pant-plazo, and several other matching accessories across multiple varieties.

The key business problems identified are: (1) the absence of a structured method to evaluate which product categories generate the most revenue and (2) the inability to track customer demand patterns across product categories, which leads to inefficient inventory planning and stock management decisions.

To address these problems, day-wise sales and procurement data for the period January 2026 to April 2026 was collected through direct field visits and physical registers. This data will be structured and evaluated using quantitative data analysis-based methods—specifically ABC analysis, inventory turnover ratios, descriptive statistics, and moving average forecasting—using Microsoft Excel and Google Colab (Python). The expected outcome is a set of data-driven operational insights to optimize working capital cycles, balance stocking levels, and minimize overall inventory holding expenditures.

2 Organization Background

Matching Point is a B2C retail clothing and household utility store located on Powerhouse Road, Maripur, Muzaffarpur, Bihar, situated in front of Trome Hospital. The store was established on 19 May 2014 and has been operating in the local retail market for more than ten years. The store is open from 9:00 AM to 9:00 PM on all working days and is managed by its owner along with one to two supporting staff members.

The store offers a diverse product range including lungi, undergarments, sarees, blouses, suit pieces, matching dupattas, saaya (petticoat), leggings, plazo, pant-plazo, mosquito nets, and various matching accessories. Each product category includes multiple varieties to cater to different customer preferences. The business primarily serves customers from nearby residential and commercial areas through direct walk-in retail sales. Product selection and inventory decisions are currently based on the owner's personal experience and market observation rather than any systematic data analysis.

Matching Point operates strictly in the B2C segment, employing 2 staff members, with standard operational hours from 9:00 AM to 9:00 PM, catering to local retail consumers and generating an estimated annual revenue of approximately ₹10,00,000 to ₹12,00,000.

3 Problem Statement

- 3.1 Reducing Locked Working Capital: Matching Point lacks a structured data framework to track category-wise sales performance. Consequently, slow-moving garment stock remains undetected on shelves for long periods, leading to locked working capital and artificial liquidity constraints for operating costs.
- 3.2 Minimizing Total Inventory Cost: Stocking and procurement decisions are heavily dependent on manual intuition and memory rather than quantitative analysis. This lack of demand visibility causes structural inventory imbalances, creating overstocking in low-demand lines (which increases holding costs) and frequent stockouts in high-demand items (which results in lost sales revenue).

Objectives

- To reduce the locked working capital at Matching Point.
- To minimize total inventory cost at Matching Point.

4 Background of the Problem

The core operational challenge at Matching Point stems from its heavy reliance on memory-based, non-digitized records, which leads to limited financial visibility and sub-optimal stock control.

Internal Problems

Without precise, product-wise sales tracking, slow-moving garment categories like premium sarees or seasonal suit pieces remain on shelves significantly longer than anticipated. This stagnation causes a substantial amount of the store's working capital to become locked up in inactive inventory, creating artificial liquidity constraints. Since the business currently lacks a data-backed discounting or promotional mechanism to systematically liquidate this slow-moving stock, managing everyday cash flow becomes inefficient.

External Problems

The garment retail sector experiences seasonal demand fluctuations, making it difficult for small retailers to anticipate customer buying patterns without structured data. Competitive pressure in the local Muzaffarpur market further demands that the store maintain a wide product variety, which increases the risk of overstocking low-demand items while understocking high-demand categories.

5 Problem Solving Approach

5.1 Intended Data Collection with Justification

The data collection process will focus on three main areas relevant to both problem statements: sales, purchase/procurement, and store operations. Day-wise sales data, including transaction date, product category, quantity sold, and selling price per unit, will be collected from daily sales bills and physical registers maintained by the store, since digital records are not kept. This data will primarily support the ABC and turnover analysis for Problem 1, as well as the demand pattern analysis for Problem 2. Purchase and procurement data, including cost price per category, order dates, and stock replenishment frequency, will additionally be gathered to support the turnover and cost-related aspects of Problem 1.

Store metadata such as operational timings, staff count, and estimated annual revenue will also be recorded to provide overall business context. Primary data collection was chosen over secondary sources because no external dataset can accurately capture Matching Point's specific sales and demand behavior; structured day-wise records ensure the reliability required for meaningful quantitative analysis of both problems.

5.2 Methods Used with Justification

For Problem 1 (Reducing Locked Working Capital):

- **ABC Analysis:** This quantitative method will be used to classify product categories based on their revenue contribution. The rationale is to objectively separate key revenue-generating items from underperforming ones to aid capital reallocation.
- **Inventory Turnover Analysis:** This method will be applied to calculate the velocity of stock cycles across product groups. The rationale is to isolate specific low-turnover garment categories that are contributing the most to stagnant capital.

For Problem 2 (Minimizing Total Inventory Cost):

- **Descriptive Statistical Analysis:** This method will be used to analyze parameters like mean and variance in purchasing distributions. The rationale is to rank categories based on consistent frequency and identify variation across trading periods.
- **Moving Average-based Demand Forecasting:** This time-series statistical technique will be used to smooth random variations in transactional records. The rationale is to estimate near-future demand trends to drive a structured procurement cycle.

5.3 Analysis Tools with Justification

- Microsoft Excel — This is the primary tool for this project. It will be used to organize raw data, calculate category-wise sales totals, averages, and percentages, and prepare charts and graphs for visual representation. Excel is chosen because it is simple to use, widely available, and sufficient for the volume and type of data collected from this small retail business.
- Google Colab (Python) — This will be used as a supplementary tool for any additional calculations or data visualizations that require more flexibility than Excel. It is free, easy to access, and supports libraries like Pandas and Matplotlib which can help in presenting findings more clearly.
- Charts and Graphical Representations — Categories and demand patterns will be mapped using distribution charts and comparative column charts. The rationale for choosing these specific visuals is to track variations between peak periods and off-peak trading intervals, rather than plotting standalone low-value bars or standard Pareto charts.

6 Expected Timeline

6.1 Workbreakdown Structure

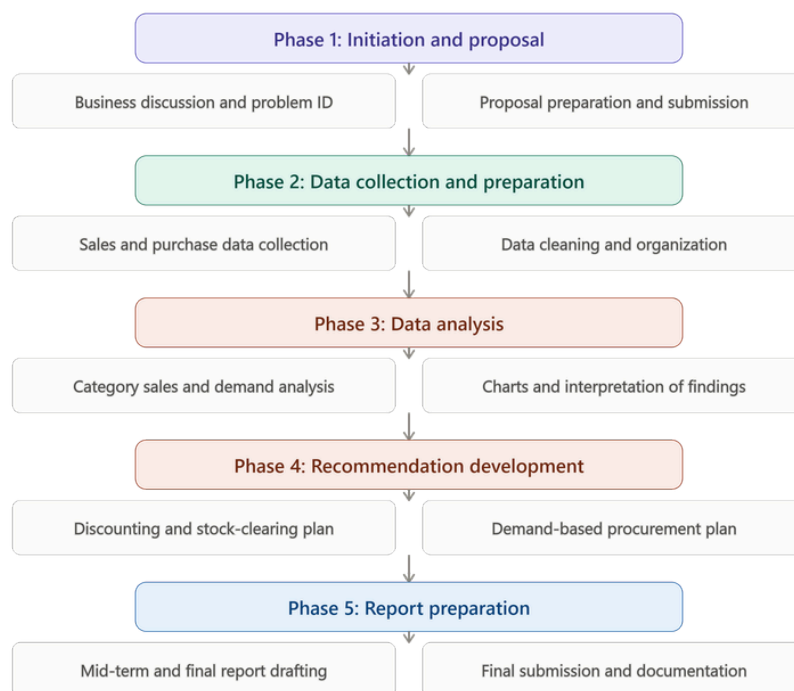


Figure 1: Work Breakdown Structure for Project Phases

6.2 Gantt Chart



Figure 2: Gantt Chart

7 Expected Outcomes

The proposed study is expected to deliver actionable, data-driven insights to optimize inventory planning and retail operations at Matching Point:

- For Problem Statement 1 (Working Capital): The analysis will identify specific slow-moving garment categories that contribute to locked working capital. The resulting data-backed discounting and stock-clearing recommendations will directly help reduce inactive inventory, improve turnover rates, and enhance daily cash flow management.
- For Problem Statement 2 (Inventory Cost): The study will reveal granular customer demand patterns and seasonal fluctuations. This supports a structured, demand-based procurement strategy that balances stocking levels, minimizes holding costs, prevents overstocking, and reduces sudden stockouts.
- Operational Impact: By transitioning from intuition-based choices to quantitative, evidence-based decision-making, the project will improve overall operational efficiency, optimize inventory investments, reduce financial wastage, and enhance store profitability.